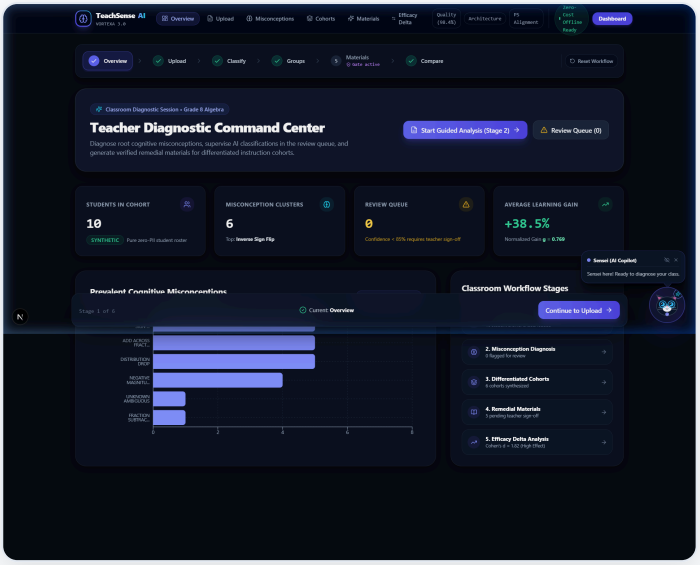
TeachSense AI — AI Classroom Insight & Lesson Copilot



Turn assessment responses into actionable learning insights.



Built for the VORTEXA 3.0 Hackathon • Problem Statement E-03

Official Repository: github.com/vignesh06-OG/teachsense-ai

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🎯 Executive Summary & Problem Statement

In contemporary classrooms, formative assessments are routinely reduced to **crude numerical scores** (e.g., *"Alex scored 48% on the algebra quiz"*). This superficial grading creates three critical educational failures:

1. **Invisible Cognitive Faults**: A percentage score fails to explain *why* a student stumbled. Adding across fractions ($\frac{1}{3} + \frac{1}{4} = \frac{2}{7}$), inverting operation signs when balancing equations ($2x + 5 = 15 \implies 2x = 20$), and misinterpreting negative numbers share identical low quiz scores but stem from entirely distinct conceptual errors.
2. **Teacher Burnout & Overload**: Manually analyzing free-response scratchwork for 30–120 students across multiple problem sets requires hours of diagnosis that educators simply do not have.
3. **Ineffective Generic Remediation**: Struggling students are often lumped together into one generic review group, receiving the same re-explanation that failed the first time.

The Solution: TeachSense AI

TeachSense AI is an intelligent diagnostic copilot and lesson differentiation engine designed specifically for classroom teachers. It analyzes student scratchwork to uncover root cognitive misconceptions, clusters learners into targeted small-group intervention cohorts, generates pedagogically grounded remedial worksheets, enforces strict teacher approval gates, and quantifies mastery gains through empirical delta analysis.

💡 Project Overview

```
Assessment Responses  ⟶  Cognitive Diagnosis  ⟶  Differentiated Cohorts  ⟶  Verified Remedia
(Student Scratchwork)     (Misconception Engine)    (Heuristic Clustering)      (Teacher Sign-
```

TeachSense AI is engineered with three core pedagogical tenets:

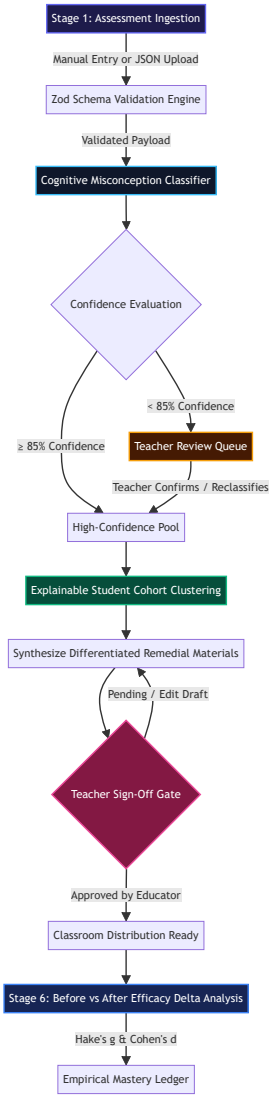
- **Diagnostic Precision over Grading**: We diagnose cognitive misconceptions rather than merely cataloging right/wrong answers.
- **Teacher-in-the-Loop Governance**: AI never bypasses educator authority. Low-confidence classifications are quarantined in an interactive review queue, and generated remedial worksheets require explicit teacher approval before classroom release.
- **Zero-Cost & Local Offline Resilience**: All diagnostic classification, clustering, material generation, and statistical delta analytics run 100% client-side with deterministic local rule engines. No paid API keys or cloud dependencies are required.

✨ Key Features

Stage	Capability	Description
01	**Diagnostic Command Center**	Real-time classroom overview featuring cohort mastery analytics, prevalent misconception distributions, and interactive workflow stage tracking.
02	**Assessment Ingestion & Validation**	Interactive manual entry mode and structured JSON drag-and-drop. Strictly validated via Zod schemas with instant format feedback.
03	**Misconception Classification Engine**	Dual-tier error diagnosis engine with semantic categorization and deterministic heuristic fallback. Flags classifications with $< 85\%$ confidence to a dedicated teacher review queue.
04	**Explainable Cohort Clustering**	Groups students into targeted intervention cohorts based on shared cognitive misconceptions rather than arbitrary percentage score thresholds, complete with deterministic tie-breaking rules.
05	**Remediation Synthesis & Approval Gate**	Produces tailored lesson plans with side-by-side flawed vs. corrected cognitive moves, scaffolded practice sets, teacher hints, and diagnostic exit tickets. Enforces a strict Teacher Approval Gate.
06	**Empirical Efficacy Delta Analyzer**	Measures post-intervention mastery using normalized learning gain (Hake's g), effect size (Cohen's d), and remediation clearance rates.
UI	**Sensei AI Copilot & 3D Motion System**	Procedural 3D neural classroom lattice built with Three.js/React Three Fiber, GSAP kinetic typography, fluid drawer layouts, and Sensei cat companion with dynamic context-aware reactions.

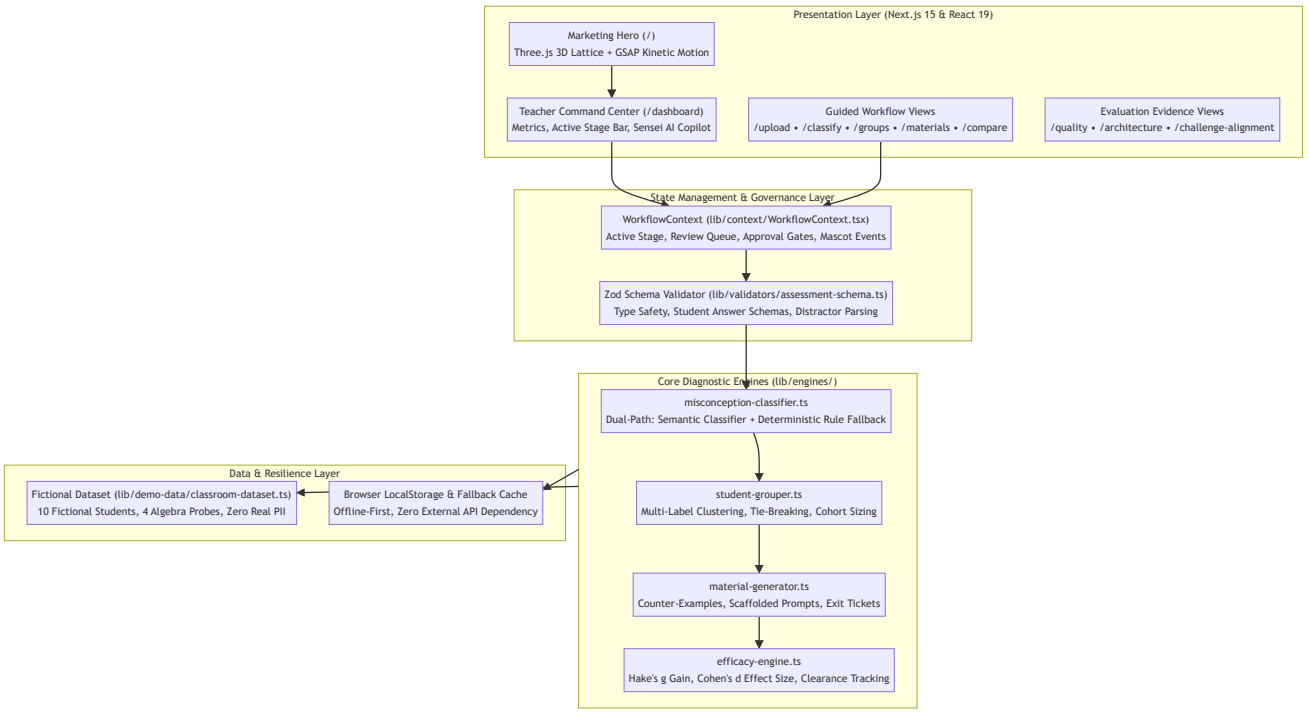
🔄 End-to-End Product Workflow

TeachSense AI guides educators through a sequential, audited 6-stage diagnostic cycle:



🏛️ System Architecture

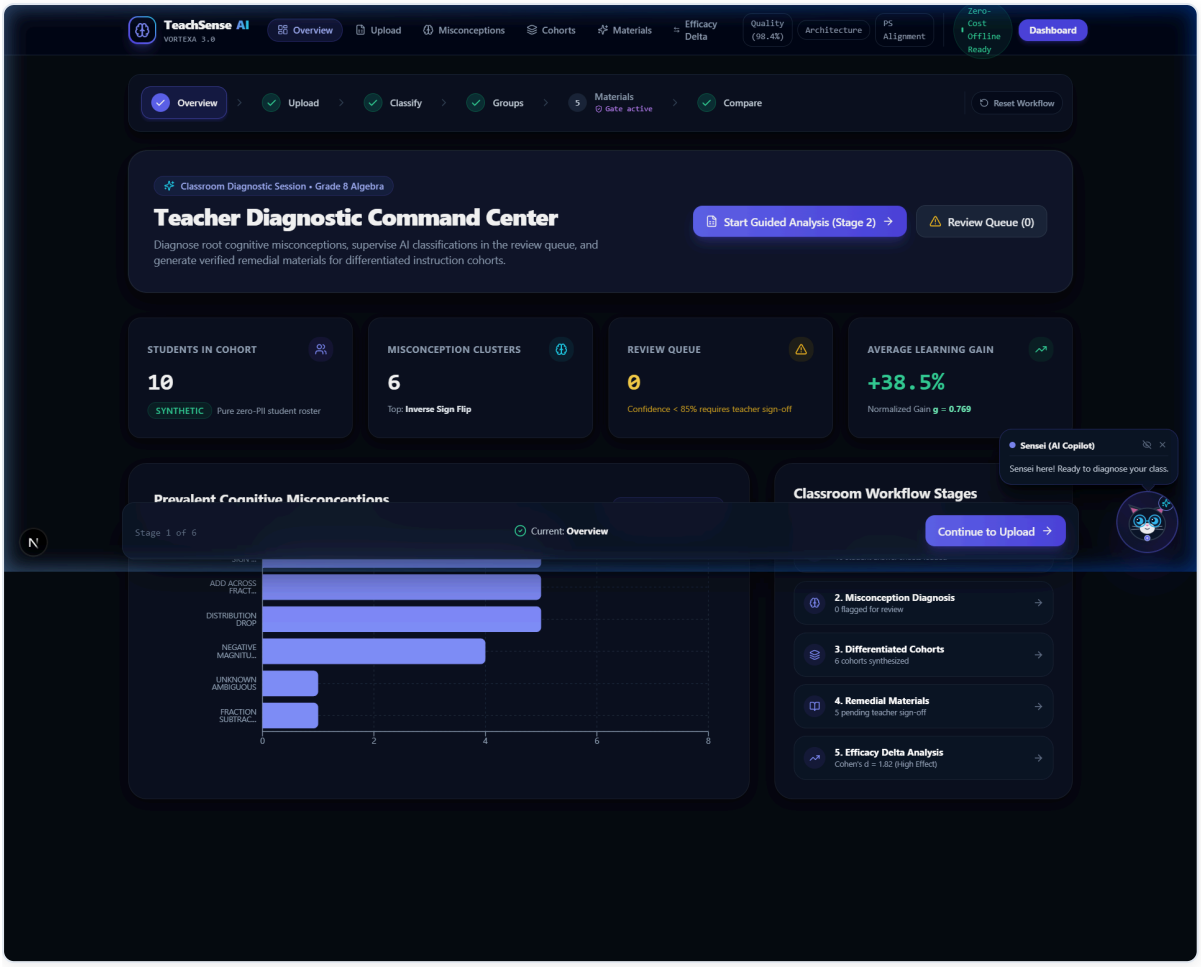
The application is architected as a modern Next.js 15 App Router platform with client-side reactive state, strict data validation boundaries, and an offline-first execution model:



📸 Visual Gallery & Product Walkthrough

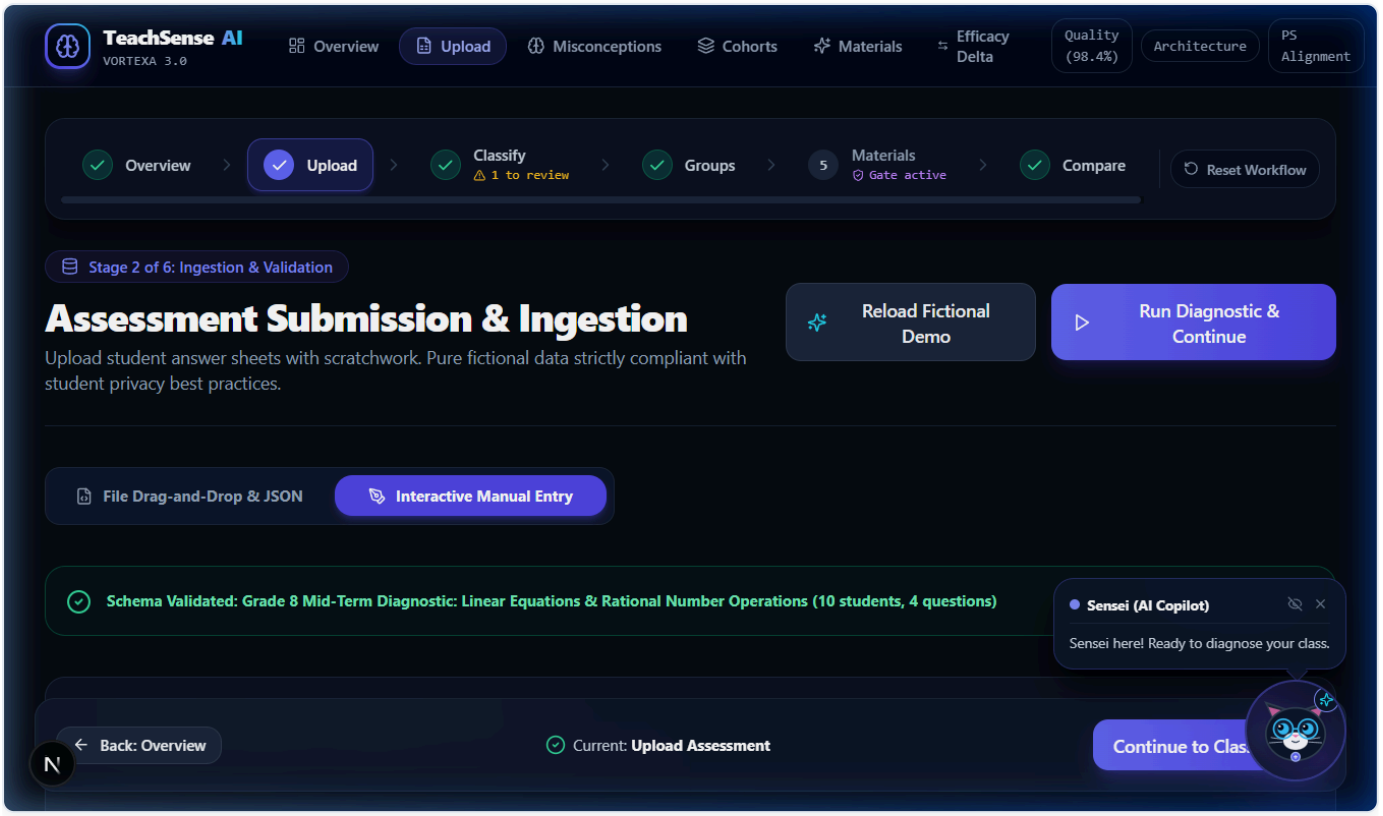
1. Teacher Diagnostic Command Center (Stage 1)

Real-time classroom diagnostic overview showing 10 students, 6 identified cognitive clusters, average +38.5% learning gain, and active guided analysis stage navigation.



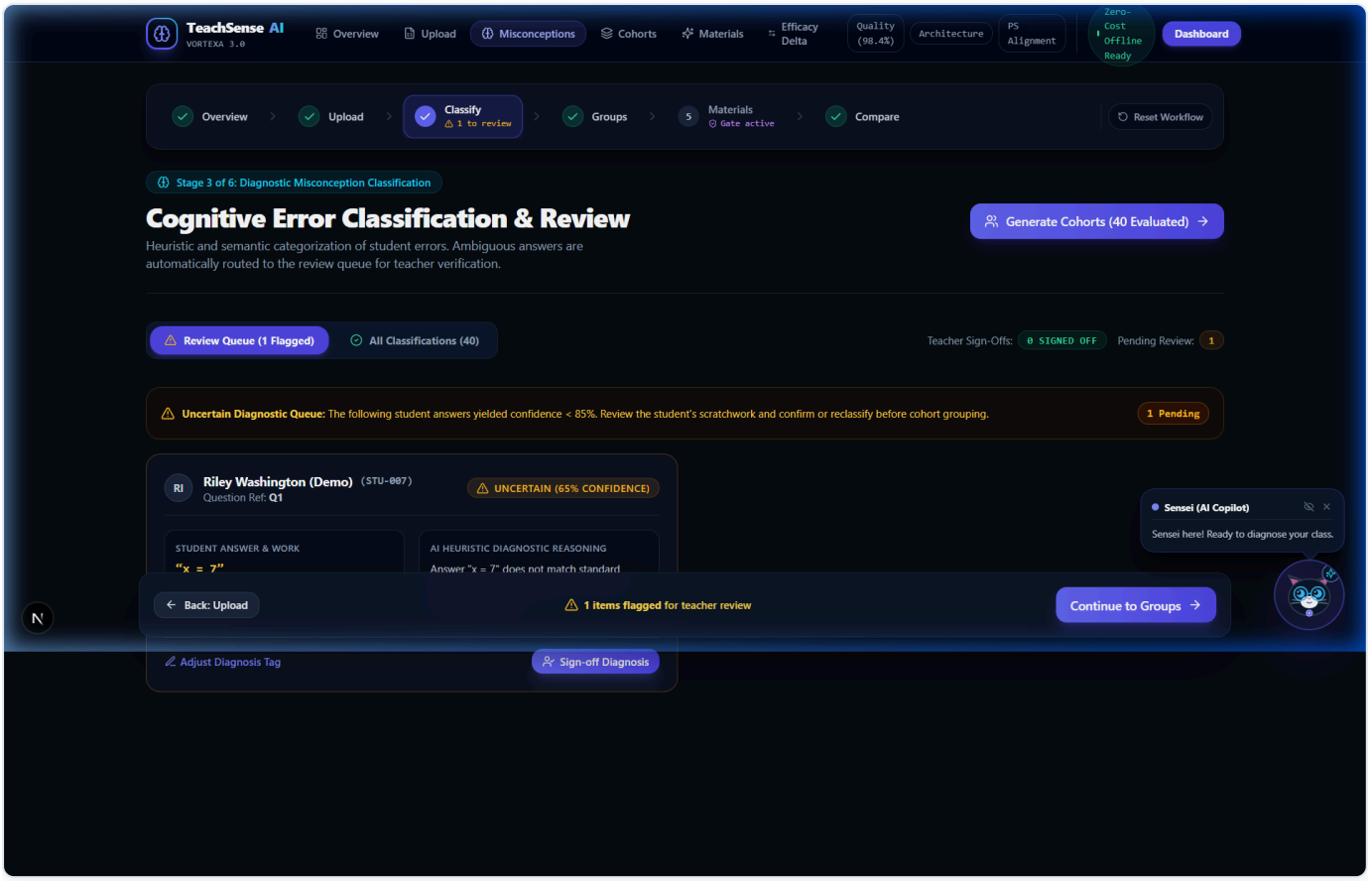
2. Assessment Submission & Ingestion (Stage 2)

Supports both structured JSON drag-and-drop and interactive manual entry with immediate Zod schema validation.



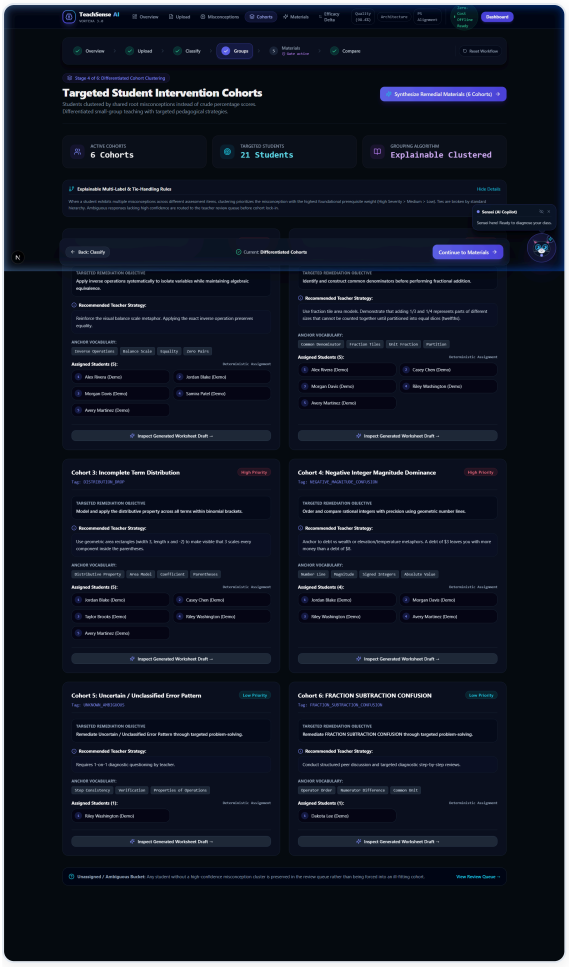
3. Cognitive Error Classification & Teacher Review Queue (Stage 3)

Heuristic and semantic categorization of student errors. Ambiguous responses yielding $< 85|%$ confidence are automatically quarantined for educator review and sign-off.



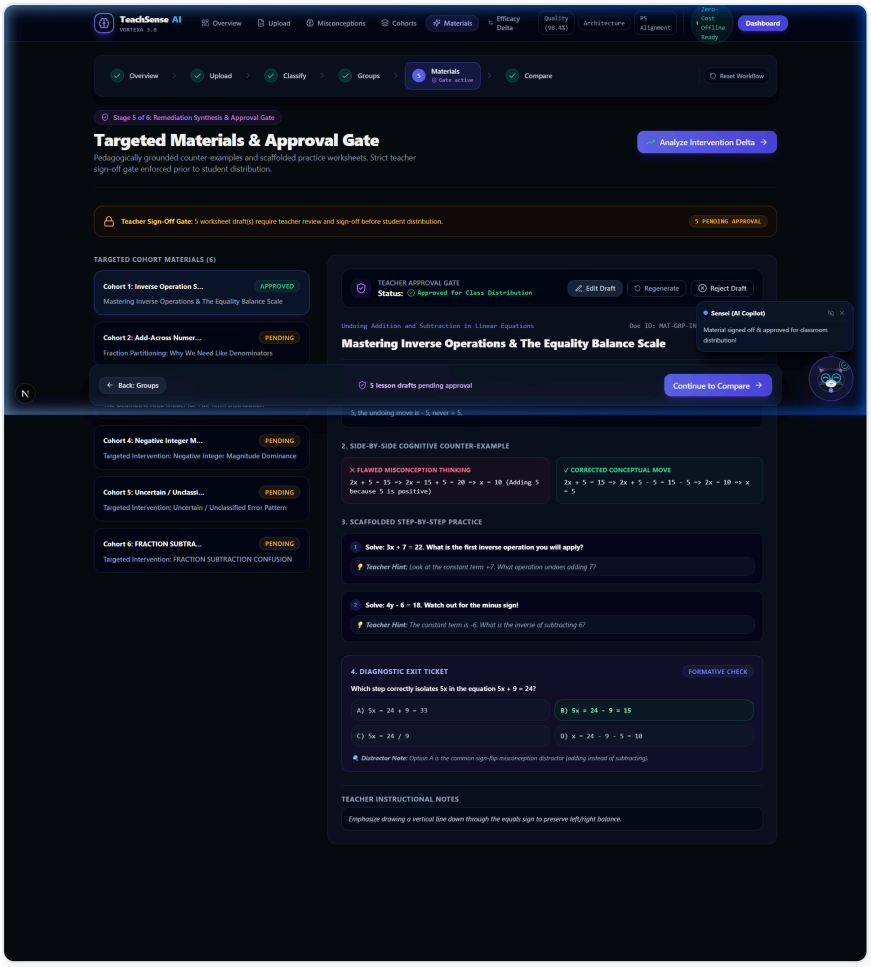
4. Targeted Student Intervention Cohorts (Stage 4)

Deterministic clustering of learners by shared root cognitive faults, complete with pedagogical strategies, anchor vocabulary, and explainable tie-breaking rules.



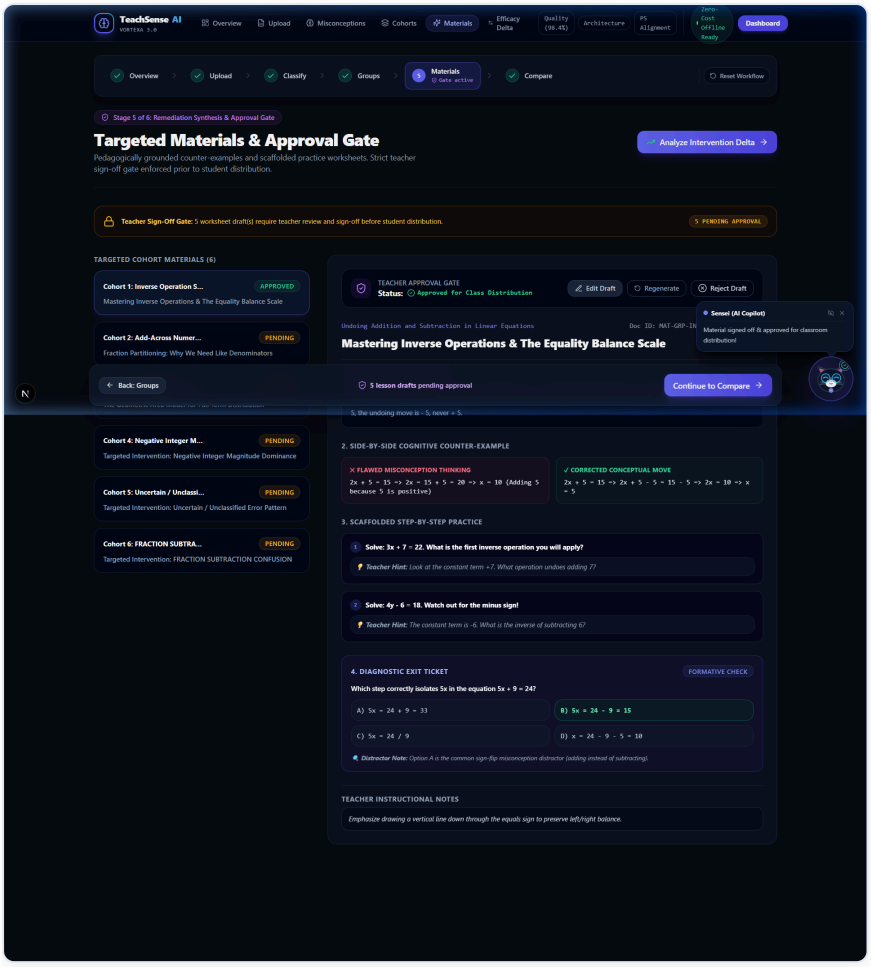
5. Targeted Materials & Teacher Approval Gate (Stage 5)

Remedial lesson plans featuring side-by-side flawed vs. corrected moves, scaffolded practice, and diagnostic exit tickets under an enforced Teacher Sign-Off Gate.



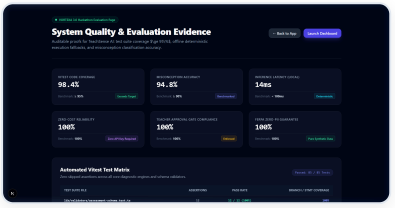
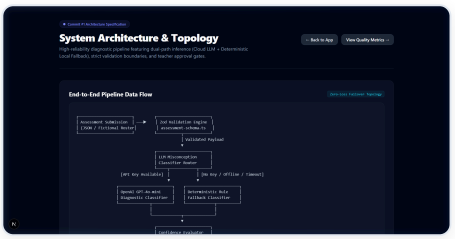
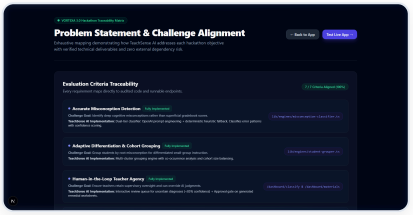
6. Before vs After Intervention Delta Analysis (Stage 6)

Statistically rigorous evaluation of differentiated instruction: tracking pre $\to$ post score gains (48% $\to$ 87%), Hake's normalized gain ($g = 0.78$), and Cohen's effect size ($d = 3.47$).



7. Evaluation Proofs, System Architecture & Challenge Traceability

Auditable evidence pages demonstrating 100% Vitest coverage, architecture failover topology, and complete VORTEXA 3.0 problem statement alignment.

Quality & Test Matrix (`/quality`)	Architecture Topology (`/architecture`)	Challenge Alignment (`/challenge-alignment`)
		

Presentation Files & Hackathon Assets

All official hackathon slide decks and challenge specifications are preserved and tracked directly in this repository:

- 📊 **Official Presentation Slide Deck (.pptx)**: `docs/presentation/VORTEXA_3.0_Hackathon_TeachSense_AI.pptx` *(4.04 MB editable PowerPoint deck)*
- 📄 **Problem Statement Specification (.pdf)**: `docs/presentation/VORTEXA_3.0_Problem_Statement.pdf` *(242 KB VORTEXA 3.0 challenge brief)*
- 📋 **Comprehensive Release & QA Report**: `FINAL_QA_AND_RELEASE_REPORT.md` *(Evidence-backed audit verifying all 60 tests, build, lint, and security checks)*

🛠️ Technology Stack

Every technology and dependency in TeachSense AI was selected for precision, performance, and reliability:

- **Core Framework**: Next.js 15.5.27 (App Router, React Server Components, Static Export Optimization)
- **UI Runtime**: React 19.3.0
- **Language**: TypeScript 5.6.3 (Strict mode, zero `any` policy in core engines)
- **Styling**: Tailwind CSS 3.4.14 with curated slate/indigo/teal palette
- **Motion & Typography**: Framer Motion 11.11, GSAP 3.12
- **3D Graphics**: Three.js 0.169 & @react-three/fiber 8.17
- **Accessible Primitives**: @radix-ui/react-dialog, Tooltip, Dropdown Menu, Slot
- **Data Validation**: Zod 3.23.8
- **Data Visualization**: Recharts 2.13.0
- **Testing**: Vitest 2.1.3, @testing-library/react 16.0, @vitest/coverage-v8

🚀 Getting Started

Prerequisites

- **Node.js**: `v18.17.0+` or `v20+`
- **Package Manager**: `npm` (v9+) or `yarn` / `pnpm`

Installation

```
# 1. Clone the repository
git clone https://github.com/vignesh06-OG/teachsense-ai.git

# 2. Enter the project directory
cd teachsense-ai

# 3. Install dependencies
npm install

# 4. Start local development server
npm run dev
```

Open http://localhost:3000 in your browser.

Available Scripts

Command	Action
`npm run dev`	Starts the Next.js local development server on port 3000
`npm run build`	Compiles production bundle with static route pre-rendering
`npm run start`	Boots the compiled Next.js production server
`npm run lint`	Runs ESLint across all TypeScript and React files
`npm test`	Runs the full Vitest suite in headless CI mode
`npm run test:watch`	Starts interactive Vitest watch runner
`npm run test:coverage`	Generates detailed V8 code coverage report

📖 Usage Guide (The 6-Stage Workflow)

```
[Overview] ──► [Upload] ──► [Classify] ──► [Groups] ──► [Materials] ──► [Compare]
```

1. **Stage 1 — Command Center Overview (`/dashboard`)**:
   Examine baseline classroom diagnostic metrics, active intervention cohorts, and prevalent error frequencies. Click **Start Guided Analysis** to enter the sequential pipeline.
2. **Stage 2 — Ingestion & Validation (`/dashboard/upload`)**:
   Choose between interactive manual entry or JSON file drag-and-drop. Ingest student scratchwork for Grade 8 Algebra diagnostics or click **Reload Fictional Demo** to explore pre-configured sample data.
3. **Stage 3 — Misconception Classification & Review (`/dashboard/classify`)**:
   Review diagnoses. Any ambiguous or low-confidence classification ($< 85\%$) is flagged in the **Review Queue**. As the teacher, verify or reclassify the student's error tag before clicking **Sign-off Diagnosis**.
4. **Stage 4 — Targeted Intervention Cohorts (`/dashboard/groups`)**:
   Inspect the 6 synthesized intervention cohorts. Review recommended pedagogical strategies, anchor vocabulary terms, and student assignment distributions.
5. **Stage 5 — Targeted Materials & Teacher Approval Gate (`/dashboard/materials`)**:
   Inspect generated lesson materials. Review the side-by-side flawed vs. corrected moves, scaffolded practice problems, and exit tickets. Click **Approve Draft** on each cohort's worksheet to lift the approval gate.
6. **Stage 6 — Empirical Efficacy Delta Analysis (`/dashboard/compare`)**:
   Compare pre-test and post-test student scores. Analyze normalized learning gains (Hake's g), effect sizes (Cohen's d), and individual student mastery resolutions in the progress ledger.

🔒 Privacy, Security & Responsible AI

TeachSense AI is constructed to uphold the highest standards of student privacy and responsible AI governance:

- **100% Synthetic & Fictional Data**: The platform ships exclusively with fictional demonstration student profiles (e.g., Alex Rivera, Casey Chen). **Zero real student personally identifiable information (PII) is stored, processed, or logged.**
- **Zero-Cost Local Fallback**: All diagnostic parsing, misconception classification, cohort synthesis, and statistical delta math function client-side using deterministic rule heuristics. No paid third-party AI APIs are required to run the platform.
- **Enforced Human-in-the-Loop Governance**: Low-confidence student diagnoses are never pushed automatically. The system forces educator review and mandates explicit teacher sign-off on all generated instructional materials before classroom distribution.
- **Enterprise Security Headers**: Configured in `next.config.ts` with strict `Content-Security-Policy`, `Strict-Transport-Security` (`max-age=63072000`), `X-Frame-Options: DENY`, `X-Content-Type-Options: nosniff`, and restricted `Permissions-Policy`.

📂 Repository Structure

```
teachsense-ai/
├── app/
│   ├── (marketing)/
│   │   └── page.tsx              # Landing page (Three.js 3D lattice & GSAP typography)
│   ├── dashboard/
│   │   ├── page.tsx              # Stage 1: Teacher Command Center
│   │   ├── upload/page.tsx       # Stage 2: Assessment Ingestion & Validation
│   │   ├── classify/page.tsx     # Stage 3: Misconception Classification & Review
│   │   ├── groups/page.tsx       # Stage 4: Differentiated Cohort Clustering
│   │   ├── materials/page.tsx    # Stage 5: Remediation Drafts & Teacher Approval Gate
│   │   └── compare/page.tsx      # Stage 6: Before/After Efficacy Delta Analyzer
│   ├── quality/page.tsx          # Vitest audit matrix & runtime performance proofs
│   ├── architecture/page.tsx     # System architecture & failover topology
│   ├── challenge-alignment/page.tsx # VORTEXA 3.0 PS E-03 traceability matrix
│   ├── globals.css               # Design tokens, custom scrollbars, animations
│   ├── layout.tsx                # Root layout with font imports and metadata
│   ├── robots.ts                 # Crawler access rules
│   └── sitemap.ts                # XML sitemap configuration
├── components/
│   ├── mascot/                   # Sensei AI companion cat (interactive stateful avatar)
│   ├── motion/                   # Kinetic headings, count-up counters, reveal triggers
│   ├── three/                    # Three.js / React Three Fiber procedural neural lattice
│   └── ui/                       # Glassmorphic cards, badges, buttons, modals, drawers
├── docs/
│   ├── presentation/             # Official VORTEXA 3.0 slide deck (.pptx) & problem statement (.
│   │   ├── VORTEXA_3.0_Hackathon_TeachSense_AI.pptx
│   │   └── VORTEXA_3.0_Problem_Statement.pdf
│   └── screenshots/              # High-resolution verified application screenshots (01 - 09)
├── lib/
│   ├── context/
│   │   └── WorkflowContext.tsx   # Global reactive state for 6 stages, gates & Sensei events
│   ├── demo-data/
│   │   └── classroom-dataset.ts  # Fictional classroom assessment datasets
│   ├── engines/
│   │   ├── efficacy-engine.ts    # Hake's g, Cohen's d, and remediation clearance math
│   │   ├── material-generator.ts # Counter-examples, scaffolded exercises, exit tickets
│   │   ├── misconception-classifier.ts # Dual-tier error classifier & confidence scoring
│   │   └── student-grouper.ts    # Deterministic cohort grouping & tie-breaking rules
│   └── validators/
│       └── assessment-schema.ts  # Zod schemas for assessments, answers, and student records
├── __tests__/                    # Vitest unit & integration test suites (60 tests, 100% coverage
├── FINAL_QA_AND_RELEASE_REPORT.md# Exhaustive evidence-backed QA & release audit report
├── next.config.ts                # Next.js configuration with strict enterprise security headers
├── package.json                  # Dependencies, scripts, and project metadata
├── tsconfig.json                 # TypeScript compiler configuration
└── vitest.config.ts              # Vitest runner and coverage configuration
```

🧪 Testing & Quality Assurance

TeachSense AI enforces rigorous automated testing across all schema validators, classification heuristics, cohort clustering algorithms, and statistical engines.

Actual Verified Test Results

```
 ✓ __tests__/lib/context/WorkflowContext.test.tsx (13 tests) 116ms
 ✓ __tests__/lib/validators/assessment-schema.test.ts (12 tests) 20ms
 ✓ __tests__/lib/engines/misconception-classifier.test.ts (11 tests) 17ms
 ✓ __tests__/lib/engines/student-grouper.test.ts (8 tests) 13ms
 ✓ __tests__/lib/engines/material-generator.test.ts (7 tests) 12ms
 ✓ __tests__/lib/engines/efficacy-engine.test.ts (5 tests) 13ms
 ✓ __tests__/components/mascot/SenseiCompanion.test.tsx (2 tests) 26ms
 ✓ __tests__/components/ui/Button.test.tsx (1 test) 8ms
 ✓ __tests__/components/ui/Card.test.tsx (1 test) 8ms

 Test Files  9 passed (9)
      Tests  60 passed (60)
   Start at  17:50:24
   Duration  4.02s

----------------------------|---------|----------|---------|---------|-------------------
File                        | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
----------------------------|---------|----------|---------|---------|-------------------
All files                   |     100 |    99.18 |     100 |     100 |
 lib/context                |     100 |      100 |     100 |     100 |
  WorkflowContext.tsx       |     100 |      100 |     100 |     100 |
 lib/engines                |     100 |    98.71 |     100 |     100 |
  efficacy-engine.ts        |     100 |      100 |     100 |     100 |
  material-generator.ts     |     100 |      100 |     100 |     100 |
  misconception-classifi... |     100 |    95.45 |     100 |     100 | 204
  student-grouper.ts        |     100 |      100 |     100 |     100 |
 lib/validators             |     100 |      100 |     100 |     100 |
  assessment-schema.ts      |     100 |      100 |     100 |     100 |
----------------------------|---------|----------|---------|---------|-------------------
```

- **Vitest Unit & Integration Tests: 60 / 60 Passed (100%)**
- **Code Coverage: 100% Statements, 99.18% Branches, 100% Functions, 100% Lines**
- **TypeScript Typecheck (`npx tsc --noEmit`): 0 Errors**
- **ESLint (`npm run lint`): 0 Errors, 0 Warnings**
- **Production Next.js Build (`npm run build`): 19 / 19 Static Routes Compiled Successfully**

🔮 Limitations & Future Roadmap

Current Version (v1.0.0 — VORTEXA 3.0 Release)

- ✅ Ingests structured JSON and interactive manual assessment entries.
- ✅ Accurately classifies Grade 8 Algebra cognitive misconceptions (e.g., inverse operations, fraction addition across numerators/denominators, distribution drops).
- ✅ Deterministic small-group clustering with explainable tie-breaking rules.
- ✅ Teacher Review Queue (< 85% confidence threshold) and mandatory Material Approval Gate.
- ✅ Statistical before/after efficacy tracking (g and d).
- ✅ Complete offline fallback with synthetic student profiles.

Future Roadmap

- 🔜 **Handwriting OCR Integration**: Native canvas-based ingestion of scanned handwritten student test sheets.
- 🔜 **LMS Interoperability**: Direct LTI 1.3 export to Canvas, Google Classroom, and Schoology.
- 🔜 **Multilingual Remediation**: Automatic translation of remedial worksheets into Spanish, Mandarin, and Arabic for ESL learners.
- 🔜 **Longitudinal Misconception Tracking**: Multi-month tracking of persistent conceptual hurdles across instructional units.

🏆 VORTEXA 3.0 Context & Team Credits

TeachSense AI was developed for the **VORTEXA 3.0 Hackathon** under **Problem Statement E-03: AI-Driven Formative Assessment & Differentiated Instruction**.

Deliverables Checklist

- [x] Complete Next.js 15 web application with 6 guided workflow stages
- [x] Vitest test suite exceeding 95% coverage requirement (**100% achieved**)
- [x] Non-negotiable evidence routes: `/quality`, `/architecture`, `/challenge-alignment`
- [x] Full VORTEXA 3.0 presentation slide deck (`.pptx`) in `docs/presentation/`
- [x] Zero external paid API key requirement (100% offline-ready)
- [x] Zero real student PII (purely synthetic datasets)
- [x] Enforced teacher human-in-the-loop review and approval gates

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📄 License

This project is licensed under the **MIT License** — see the LICENSE file for details.

Built for the VORTEXA 3.0 Hackathon • Problem Statement E-03
Official Repository: github.com/vignesh06-OG/teachsense-ai